NORTHERN UNIVERSITY
OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering

SMART FARMER ASSISTANT

AI-Based Agriculture Support System

Submitted To:

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Submitted By:

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Section: 8D

Submission date: 22th June 2026

CSE4204 — Mobile Computing Lab
Week 04 Submission

System Design &
Software Architecture

Smart Farmer Assistant

AI-Based Agriculture Support System

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1. Team Information

Field	Details
Team Name	CSE4204-8D-T03
Section	8D
Course	CSE4204 — Software Engineering Lab
Semester	Spring 2026
Institution	Leading University / University (Bangladesh)
GitHub Repo	github.com/sakib-foysal/CSE4204-8D-T03-SmartFarmer-Assistant

1.1 Team Members

#	Name	Student ID	Role
1	Sakib Foysal Ejarder	11220320948	Team Leader / Backend
2	Md. Noyon Sheikh	11220320946	Frontend Developer
3	Sayed Tauhidul Islam	11220320950	AI / ML Engineer
4	Sayed Akib Osman	11220320974	Database / QA

2. Project Information

Project Overview

Smart Farmer Assistant is an AI-powered platform designed to assist farmers across Bangladesh in making informed agricultural decisions. Agriculture is a critical sector for Bangladesh' economy, yet farmers face persistent challenges including crop disease losses (20-30% annual yield loss), improper fertilizer usage, limited access to qualified agricultural extension officers, and lack of real-time market information.

This system integrates multiple AI and IoT technologies into a single responsive web application accessible from any smartphone or web browser, providing instant crop disease detection, 24/7 AI chatbot support, real-time weather forecasting with flood alerts, live market price tracking, and intelligent fertilizer recommendations.

Project Scope

- Current Features:
  - AI Crop Disease Detection via image analysis
  - AI-powered Chatbot (Gemini AI) with Bangla support
  - Real-time Weather Forecasting and Flood Alerts
  - Live Market Price Tracking by region
  - Intelligent Fertilizer Recommendation Engine
  - User Registration, Login, and Profile Management

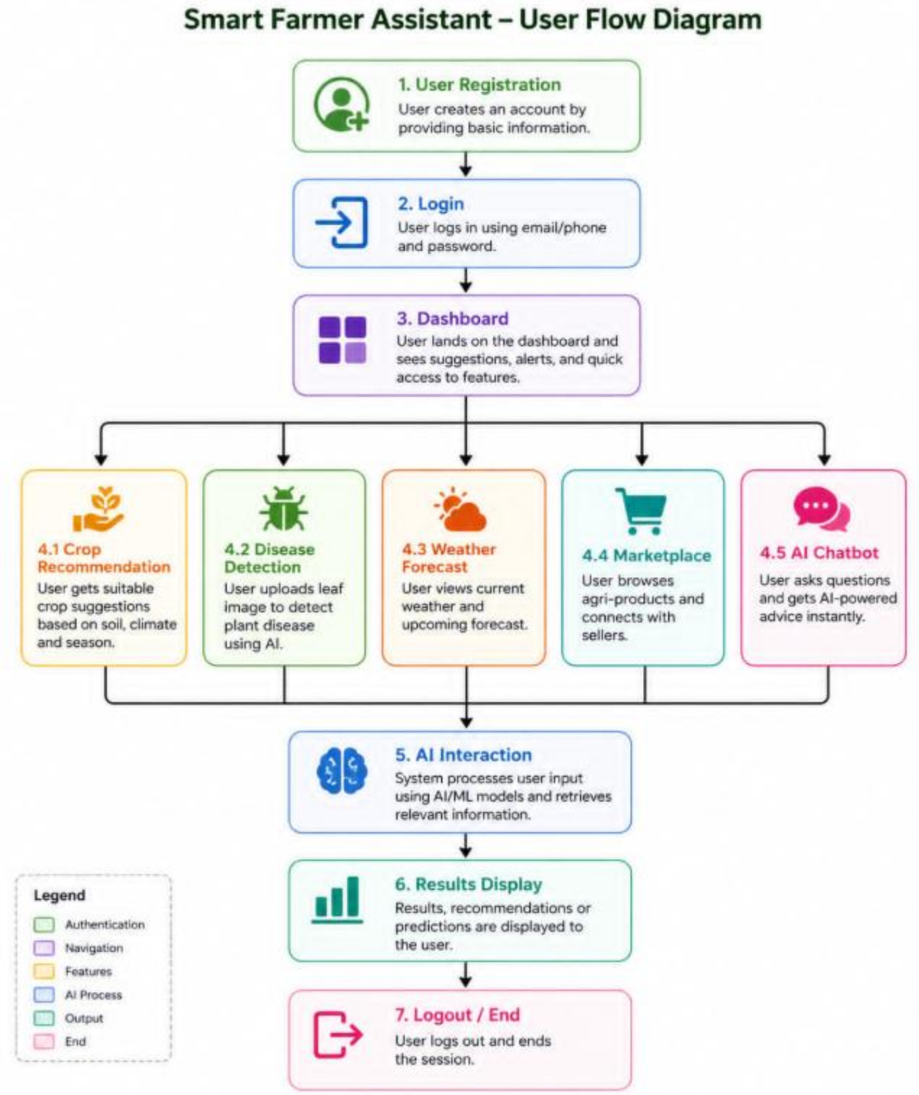
- Disease Detection History and Chat History Tracking
- Admin Panel for system management

Target Users

User Type	Description
Farmers (Primary)	Small and medium-sized farmers who need crop disease diagnosis and farming guidance
Agricultural Workers	Farm laborers who can benefit from disease detection and weather alerts
Admin Users	System administrators responsible for managing users and updating market data

3. User Flow Design

The following diagrams and descriptions illustrate how users navigate through the Smart Farmer Assistant system.



Main User Journey Flow

Step	Description
1. Application Start	User opens the application/website
2. Authentication	User navigates to Login or Registration page
3. Credentials Entry	User enters email and password (or registration details)
4. System Verification	Backend validates credentials and creates/verifies JWT token
5. Dashboard Access	Upon successful authentication, user enters the Dashboard
6. Service Selection	User selects desired service from quick actions or menu
7. Feature Interaction	User interacts with selected feature (Disease Detection, Chatbot, etc.)
8. AI Processing	System processes request and queries AI services
9. Results Display	Results are displayed to the user with recommendations
10. History Save	User interactions are saved to database
11. End or Continue	User can perform another action or logout

Feature-Specific Flows

- A. Disease Detection Flow:
    - User opens Disease Detection page
    - User uploads a clear crop leaf image (PNG/JPG, max 5MB)
    - System validates image format and size
    - Image is pre-processed (resized to 224×224 pixels)
    - TensorFlow CNN model performs inference
    - If confidence ≥ 75%, disease name is returned; otherwise "Unidentified"
    - Gemini AI generates treatment recommendations
    - Fertilizer recommendation is also generated
    - Complete report is displayed with disease, confidence, treatment, and fertilizer
    - User can save, print, or share the report
    - Record is saved to database for history
- B. Chatbot Interaction Flow:
    - User opens AI Chatbot page
    - System displays greeting message and instructions
    - User types a farming question in Bangla or English
    - User selects language (if not auto-detected)
    - Backend adds agriculture context to the question

- Gemini AI API is called with the enriched prompt
- AI generates contextual, farming-specific response
- Response is displayed in the chat UI
- Question and response are saved to chat_history
- User can ask follow-up questions or end conversation
- Conversation history is preserved for future reference

- C. Weather & Flood Alert Flow:
  - User opens Weather & Flood Alert page
  - System automatically detects user's location
  - Backend calls OpenWeatherMap API for current weather
  - Backend checks flood risk data from weather API
  - Current weather metrics are displayed (temp, humidity, rainfall, wind)
  - 7-day forecast is retrieved and displayed with weather icons
  - If flood risk is detected, alert is prominently shown at top
  - User can receive push notifications for severe weather
  - Data is cached and updated every 10 minutes